

CQC2

Calculus II

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A test driver in a newly released electric sedan records velocity every 2 seconds: 10, 14, 20 and 28 ft/s at $t = 0, 2, 4$ and 6 s. The velocity increases the whole time and the car is silver. Using a left Riemann sum, estimate the distance traveled from $t = 0$ to $t = 6$ and say whether it is an over- or underestimate.

- A. 88 ft, an overestimate.
- B. 88 ft, an underestimate.
- C. 124 ft, an overestimate.
- D. 106 ft, exact.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

2. [TRANSFER — reversed stem / different variable]

Water drains from a tank at a rate that decreases over time. Which Riemann sum is guaranteed to underestimate the total water drained?

- A. The right Riemann sum.
- B. The trapezoidal sum.
- C. The left Riemann sum.
- D. The left sum, because left sums always underestimate.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

3.

An engineer modeling heat flow in a new warehouse defines $F(x) = \int \text{from } 1 \text{ to } x^3 \text{ of } (t^2 + 1) dt$. The warehouse has 40 loading bays. What is $F'(x)$?

- A. $x^6 + 1$
- B. $6x^5$
- C. $3x^2(x^6 + 1)$
- D. $3x^2(x^6 + 1) - 2$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

4.

A delivery drone on a test flight over a new distribution center moves along a straight line with velocity $v(t) = t^2 - 4t + 3$ m/s for $0 \leq t \leq 3$. Its battery was fully charged at launch. What total distance does it travel?

- A. $8/3$ m
- B. $4/3$ m
- C. 3 m
- D. 0 m

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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5. [TRANSFER — reversed stem / different variable]

A particle has velocity $v(t) = 6t^2$ and starts at position $s(0) = 5$. What is $s(2)$?

- A. 16
- B. 29
- C. 5
- D. 21

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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6.

A student in a study group that meets in the campus library evaluates $\int$ from 0 to 2 of $x(x^2 + 1)^3 dx$ using $u = x^2 + 1$. The group has a quiz on Friday. What is the value?

- A. 156
- B. 78
- C. 312
- D. 2

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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7.

An analyst at a fast-growing startup needs $\int$ from 0 to 1 of $x \cdot e^x dx$ for a revenue model. The startup has 25 employees. What is the value?

- A. $e/2$
- B. $2e - 1$
- C. 1
- D. e

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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8. [TRANSFER — reversed stem / different variable]

To integrate $\int x \cdot \ln x \, dx$ by parts, which choice of u makes the problem simpler?

- A. $u = x \ln x, dv = dx$
- B. $u = x, dv = \ln x \, dx$
- C. Either choice always works equally well.
- D. $u = \ln x, dv = x \, dx$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A landscape architect designing a curved garden bed for a public park models the edges as $y = 4x - x^2$ and $y = x$, with x in meters. The bed will hold native plants. What is the area of the region enclosed by the two curves?

- A. $8/3 \, \text{m}^2$
- B. $4.5 \, \text{m}^2$
- C. $-4.5 \, \text{m}^2$
- D. $9 \, \text{m}^2$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A machinist making a custom part on a new lathe revolves the region between $y = x$ and $y = x^2$ for $0 \leq x \leq 1$ about the x -axis. The part will be polished steel. What is the volume?

- A. $\pi/30$
- B. $2/15$
- C. $2\pi/15$
- D. $\pi/6$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A lab culture in a newly certified biosafety room starts with 500 bacteria and doubles every 3 hours. The technician checks it at the start of each shift. How many bacteria are there after 12 hours?

- A. 8,000
- B. 2,500
- C. 4,000
- D. 2,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A population modeled by $y = y_0 e^{kt}$ doubles every 3 hours. What is k ?

- A.** $\ln 3 / 2 \approx 0.549$ per hour
- B.** $3 / \ln 2 \approx 4.33$ per hour
- C.** $2/3$ per hour
- D.** $\ln 2 / 3 \approx 0.231$ per hour

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A test driver in a ~~newly released electric sedan~~ records velocity every 2 seconds: 10, 14, 20 and 28 ft/s at $t = 0, 2, 4$ and 6 s. The velocity increases the whole time ~~and the car is silver~~. Using a left Riemann sum, estimate the distance traveled from $t = 0$ to $t = 6$ and say whether it is an over- or underestimate.

Target word: left-sum value and error direction **Decides:** Increasing function $\rightarrow$ left sum underestimates.

A	Inverted operation	Correct value, wrong direction for an increasing function.
B	Correct	$2(10 + 14 + 20) = 88$; left endpoints are the lowest values on each interval of an increasing function.
C	Confusion set	That is the right sum, $2(14 + 20 + 28)$.
D	Adjacent category	The trapezoid average of the two sums; still an estimate.

2. Answer A Concept 1 · transfer

Water drains from a tank at a rate that decreases over time. Which Riemann sum is guaranteed to underestimate the total water drained?

Target word: which sum underestimates a decreasing rate **Decides:** For decreasing functions, right endpoints are the lowest.

A	Correct	On a decreasing function each right endpoint is the smallest value on its interval.
B	Adjacent category	Its error direction depends on concavity, not on decreasing.
C	Inverted operation	Left sums overestimate a decreasing function.
D	Absolute language	True only for increasing functions.

3. Answer C Concept 2

An engineer modeling heat flow in a ~~new warehouse~~ defines $F(x) = \int$ from 1 to x^3 of $(t^2 + 1) dt$. ~~The warehouse has 40 loading bays~~. What is $F'(x)$?

Target word: $F'(x)$ **Decides:** FTC part 1 with chain rule: $f(x^3) \cdot 3x^2$.

A	Dropped element	Leaves out the chain-rule factor $3x^2$.
B	Inverted operation	Differentiates the integrand at x^3 instead of evaluating it.
C	Correct	$f(x^3) = x^6 + 1$, multiplied by the derivative of x^3 , which is $3x^2$.
D	Adjacent category	Subtracts $f(1) = 2$; the constant lower limit contributes nothing to the derivative.

4. Answer A Concept 3

A delivery drone on a test flight ~~over a new distribution center~~ moves along a straight line with velocity $v(t) = t^2 - 4t + 3$ m/s for $0 \leq t \leq 3$. ~~Its battery was fully charged at launch~~. What total distance does it travel?

Target word: total distance **Decides:** v changes sign at $t = 1$, so split the integral.

A	Correct	$\int_0^1 v = 4/3$ and $\int_1^3 v = -4/3$; distance = $4/3 + 4/3 = 8/3$.
B	Dropped element	Counts only the first interval.
C	Wrong stem number	The length of the time interval, not a distance.
D	Adjacent category	That is the displacement: $\int_0^3 v dt = 9 - 18 + 9 = 0$.

5. Answer D Concept 3 • transfer

A particle has velocity $v(t) = 6t^2$ and starts at position $s(0) = 5$. What is $s(2)$?

Target word: position with an initial value **Decides:** $s(t) = 2t^3 + 5$.

A	Dropped element	Leaves out the starting position.
B	Inverted operation	Adds $v(2) = 24$ to 5 instead of integrating velocity.
C	Wrong stem number	The starting position.
D	Correct	$s(2) = 2(8) + 5 = 21$.

6. Answer B Concept 4

A student ~~in a study group that meets in the campus library~~ evaluates $\int$ from 0 to 2 of $x(x^2 + 1)^3 dx$ using $u = x^2 + 1$. ~~The group has a quiz on Friday.~~ What is the value?

Target word: value of the integral **Decides:** $du = 2x dx$; limits become $u = 1$ to $u = 5$.

A	Dropped element	Leaves out the $1/2$ from $du = 2x dx$.
B	Correct	$(1/2)\int_1^5 u^3 du = (1/2)(625 - 1)/4 = 78$.
C	Inverted operation	Multiplies by 2 instead of $1/2$.
D	Wrong stem number	Keeps the x -limits 0 and 2: $(1/2)(16 - 0)/4 = 2$.

7. Answer C Concept 5

An analyst ~~at a fast-growing startup~~ needs $\int$ from 0 to 1 of $x \cdot e^x dx$ for a revenue model. ~~The startup has 25 employees.~~ What is the value?

Target word: value of the integral **Decides:** $u = x$, $dv = e^x dx \rightarrow xe^x - e^x$.

A	Adjacent category	Integrates each factor separately, $(x^2/2)e^x$, which is not how products integrate.
B	Inverted operation	Uses $xe^x + e^x$, dropping the minus sign in the formula.
C	Correct	$[xe^x - e^x]$ from 0 to 1 = $(e - e) - (0 - 1) = 1$.
D	Dropped element	Evaluates only $uv = xe^x$ and leaves out $-\int v du$.

8. Answer D Concept 5 • transfer

To integrate $\int x \cdot \ln x dx$ by parts, which choice of u makes the problem simpler?

Target word: choice of u **Decides:** $\ln x$ simplifies when differentiated; it has no simple antiderivative to use as v .

A	Adjacent category	Legal, but gives a new integral that is no simpler.
B	Inverted operation	Requires integrating $\ln x$ as v , making the problem harder.
C	Absolute language	The choice decides whether the new integral is easier.
D	Correct	$du = (1/x)dx$ cancels with $v = x^2/2$, leaving $\int x/2 dx$.

9. Answer B Concept 6

A landscape architect designing a curved garden bed for a public park models the edges as $y = 4x - x^2$ and $y = x$, with x in meters. ~~The bed will hold native plants.~~ What is the area of the region enclosed by the two curves?

Target word: enclosed area **Decides:** Curves meet at $x = 0$ and $x = 3$; top is $4x - x^2$.

A	Wrong stem number	Integrates $3x - x^2$ from 0 to 4, using the parabola's root instead of the intersection.
B	Correct	$\int_0^3 (3x - x^2) dx = 27/2 - 9 = 4.5$.
C	Inverted operation	Bottom minus top.
D	Dropped element	Area under the top curve only, $\int_0^3 (4x - x^2) dx$.

10. Answer C Concept 7

A machinist making a custom part ~~on a new lathe~~ revolves the region between $y = x$ and $y = x^2$ for $0 \leq x \leq 1$ about the x -axis. ~~The part will be polished steel.~~ What is the volume?

Target word: washer volume **Decides:** Outer radius x , inner radius x^2 .

A	Inverted operation	Squares the difference: $\pi \int (x - x^2)^2 dx = \pi/30$.
B	Dropped element	Leaves out the π .
C	Correct	$\pi \int_0^1 (x^2 - x^4) dx = \pi(1/3 - 1/5) = 2\pi/15$.
D	Adjacent category	π times the area of the region, not a volume of revolution.

11. Answer A Concept 8

A lab culture ~~in a newly certified biosafety room~~ starts with 500 bacteria and doubles every 3 hours. ~~The technician checks it at the start of each shift.~~ How many bacteria are there after 12 hours?

Target word: amount after 12 hours **Decides:** $12 / 3 = 4$ doublings.

A	Correct	$500 \cdot 2^4 = 8,000$.
B	Adjacent category	Linear growth: adds 500 per 3 hours.
C	Dropped element	$500 \cdot 2^3$ counts one doubling too few.
D	Inverted operation	Multiplies 500 by 4 instead of raising 2 to the 4th power.

12. Answer D Concept 8 • transfer

A population modeled by $y = y_0 e^{(kt)}$ doubles every 3 hours. What is k ?

Target word: solve for k **Decides:** $2 = e^{(3k)} \rightarrow k = \ln 2 / 3$.

A	Confusion set	Swaps the roles of the doubling factor and the time.
B	Inverted operation	Inverts the ratio.
C	Adjacent category	Treats the doubling as a linear rate.
D	Correct	Take $\ln$ of both sides of $2 = e^{(3k)}$.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	8	
Wrong stem number	A figure copied from the prompt that is not the answer.	4	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	11	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	2	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Riemann sums: over- or underestimate	For an increasing function a left sum underestimates and a right sum overestimates; for a decreasing function it is the reverse.	Assuming a left sum always underestimates.	A table of values, a left or right sum, and whether the estimate is high or low.
2	Fundamental theorem, part 1, with a chain rule	If $F(x) = \int \text{from } a \text{ to } g(x) \text{ of } f(t) dt$, then $F'(x) = f(g(x)) \cdot g'(x)$.	Substituting the upper limit but forgetting to multiply by its derivative.	A derivative of an integral whose upper limit is a function of x .
3	Displacement vs. total distance	Displacement = $\int v dt$; total distance = $\int v dt$, which requires splitting the interval where velocity changes sign. Position = initial position + $\int v dt$.	Reporting displacement when distance is asked, or dropping the initial position.	“Total distance traveled” or a given starting position.
4	u-substitution with changed limits	In a definite integral, after substituting u , change the limits to u -values (or return to x) and include the constant factor from du .	Keeping the old x -limits or dropping the constant.	An inside function whose derivative appears as a factor.
5	Integration by parts	$\int u dv = uv - \int v du$; choose u as the factor that simplifies when differentiated (logs, then algebraic, then trig or exponential).	Choosing u so the new integral is harder, or dropping the minus sign.	A product of unlike functions such as $x \cdot e^x$ or $x \cdot \ln x$.
6	Area between curves	Area = $\int (\text{top} - \text{bottom}) dx$ between the curves' intersection points.	Integrating only the top curve, reversing top and bottom, or using the wrong limits.	Two curves and “the region enclosed.”
7	Volume by washers	Revolving about the x -axis, $V = \pi \int (R^2 - r^2) dx$, where R and r are the outer and inner radii.	Squaring the difference $(R - r)^2$ instead of subtracting squares.	A region between two curves revolved about an axis.
8	Exponential growth and decay	If $dy/dt = ky$, then $y = y_0 e^{(kt)}$; with doubling time T , $y = y_0 \cdot 2^{(t/T)}$ and $k = \ln 2 / T$.	Treating doubling as adding a fixed amount, or inverting k .	Doubling time, half-life, or a rate proportional to the amount.

Course scope (public description): Calculus II is the study of the accumulation of change in relation to the area under a curve. It covers the knowledge and skills necessary to apply integral calculus of one variable and to use appropriate technology to model and solve real-life problems. Topics include antiderivatives; indefinite integrals; the substitution rule; Riemann sums; the fundamental theorem of calculus; definite integrals; acceleration, velocity, position, and initial values; integration by parts; integration by trigonometric substitution; integration by partial fractions; numerical integration; improper integration; area between curves; volumes and surface areas of revolution; arc length; work; center of mass; separable differential equations; direction fields; growth and decay problems; and sequences. Calculus I is a prerequisite for this course.